

Multiple Choice (10 marks)

1. The proportion of genes shared by first cousins is on average _____.
 - (a) $1.0/2$
 - (b) $1.0/4$
 - (c) $1.0/8$
 - (d) $1.0/16$
2. Workers in which of the following jobs still have mandatory retirements once they reach a certain age?
 - (a) Cab drivers
 - (b) Fire fighters
 - (c) Insurance regulators
 - (d) Computer programmers
3. Which of the following is most strongly associated with loss of bone density?
 - (a) Genetic factors
 - (b) Lifestyle factors like not exercising
 - (c) Hormonal factors such as loss of estrogen
 - (d) Cohort factors
4. Intergenerational transmission to offspring can occur as a result of parental exposures to _____.
 - (a) war
 - (b) natural disasters
 - (c) hunger
 - (d) all of these
5. The surveillance testing strategy associated with the least selection bias is:
 - (a) Compulsory
 - (b) Unlinked anonymous
 - (c) Voluntary anonymous
 - (d) Voluntary confidential

True / False (10 marks)

Answer True or False

6. True or False: Needle aspiration is a method of detecting breast cancer that involves exposure to radiation.
7. True or False: A cross-sectional design is the best approach for analyzing age and cohort differences simultaneously.
8. True or False: Loss of smell is very frequent in adults aged 80 and older, affecting more than 75% of this population.
9. True or False: Bile enters the digestive tract at the duodenum, where it aids in the digestion of fats.
10. True or False: A pulse may be felt on a patient's face where the superficial temporal artery crosses the maxilla.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how surprise visits and seeking advice from grandparents can enhance emotional well-being and strengthen intergenerational relationships among college students.
12. Discuss the mechanisms by which azidothymidine inhibits herpes viruses and analyze its effectiveness compared to other antiviral treatments in managing herpes infections.

End of Paper